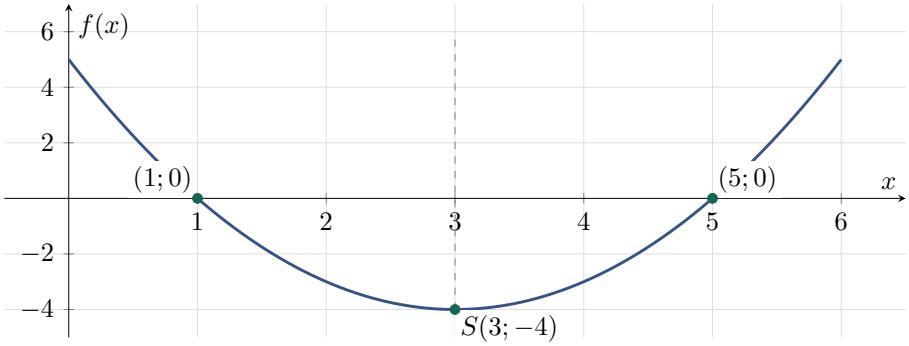
2.1 Three forms, three readings

Definition and results

A quadratic function can be written $f(x) = ax^2 + bx + c$ with $a \neq 0$. The expanded form makes it easy to compute $f(0)$; the vertex form $a(x-\alpha)^2 + \beta$ reveals vertex and extremum; a factorised form $a(x-r_1)(x-r_2)$ reveals the roots and the sign. The roots are the solutions of $f(x) = 0$.

Status and justification. Completing the square is based on $x^2+2ux = (x+u)^2-u^2$. The existence of the vertex form is proved by expanding; computing it in general is not a routine skill that is needed.

Worked example. $f(x) = x^2 - 6x + 5 = (x-3)^2 - 4 = (x-1)(x-5)$. The minimum is -4 at 3 ; the roots are 1 and 5; the y-intercept is 5.



Curve of $f(x) = (x-3)^2 - 4$. The two roots and the vertex are marked; the axis of symmetry is the line $x = 3$.

Watch out

A factorised expression is not obtained by putting each term separately in brackets.

Immediate application. Give three forms of $g(x) = x^2 + 2x - 3$. Read its roots and its minimum.

2.2 Discriminant and solving

Definition and results

For $ax^2 + bx + c = 0$ with $a \neq 0$, let $\Delta = b^2 - 4ac$. If $\Delta < 0$, no real root; if $\Delta = 0$, one root $-\frac{b}{2a}$; if $\Delta > 0$, two roots $\frac{-b - \sqrt{\Delta}}{2a}$ & $\frac{-b + \sqrt{\Delta}}{2a}$. To order the roots, take the sign of a into account.

Status and justification. Multiplying the equation by $4a \neq 0$, we obtain $(2ax + b)^2 = \Delta$. A square is positive or zero: the three cases follow. If $\Delta > 0$, $2ax + b = \pm\sqrt{\Delta}$. Splitting into cases avoids taking the square root of a negative number.

Worked example. $2x^2 + x - 3 = 0 : \Delta = 1 + 24 = 25$; $x = \frac{-1 \pm 5}{4}$ i.e. 1 and $-\frac{3}{2}$. Check: $(x - 1)(2x + 3) = 2x^2 + x - 3$.

Watch out

The denominator is $2a$, not 2 ; the sign of b is taken with its brackets in b^2.

Immediate application. Solve $x^2 - 6x + 3 = 0$ and $x^2 + 2x + 5 = 0$.

2.3 Factorising without automatically using the discriminant

Definition and results

If two real roots exist, their sum is $-\frac{b}{a}$ and their product $\frac{c}{a}$. An obvious root or a special product often gives a shorter factorisation. Polynomials of degree 2 vanishing at two distinct real numbers r and s are exactly $a(x - r)(x - s)$ with $a \neq 0$.

Status and justification. Expanding $a(x - r)(x - s) = ax^2 - a(r + s)x + ars$, identifying the coefficients gives the relations. Conversely, if r and s have this sum and this product, expanding gives back the polynomial.

Worked example. $x^2 - 7x + 12$: 3 and 4 have sum 7 and product 12. Thus $(x - 3)(x - 4)$. To also impose $f(0) = 24$, we need $12a = 24$, so $a = 2$.

Watch out

The sum of the roots is $-\frac{b}{a}$; do not forget the minus sign. A single root is not enough to determine a.

Immediate application. Determine the quadratic polynomial f such that $f(1) = f(4) = 0$ and $f(0) = -8$.

2.4 Sign, inequalities and optimisation

Definition and results

For two distinct roots $r < s$, the quadratic has the sign of a outside [r; s] and the sign opposite to a between the roots. It is zero at the roots. With no real root, it keeps the sign of a. With a double root, it has the sign of a away from it. Optimisation is done on the domain of the problem, which may limit access to the vertex.

Status and justification. The sign table of the factors $a(x - r)(x - s)$ gives the result. The vertex form compares $f(x)$ to β using the sign of $a(x - \alpha)^2$.

Worked example. $-2(x - 1)(x - 5) \geq 0$ is equivalent to $x \in [1; 5]$. For $A(x) = x(12 - x)$ on [0; 12], $A(x) = 36 - (x - 6)^2 \leq 36$, with equality at 6. A rectangle with perimeter 24 m therefore has a max area of 36 m ^{2} when it is square.

Watch out

In a strict inequality, the roots are excluded. Check the geometric bounds before concluding.

Immediate application. Solve $x^2 - 4x - 5 < 0$, then maximise $x(8 - x)$ on [0; 3].